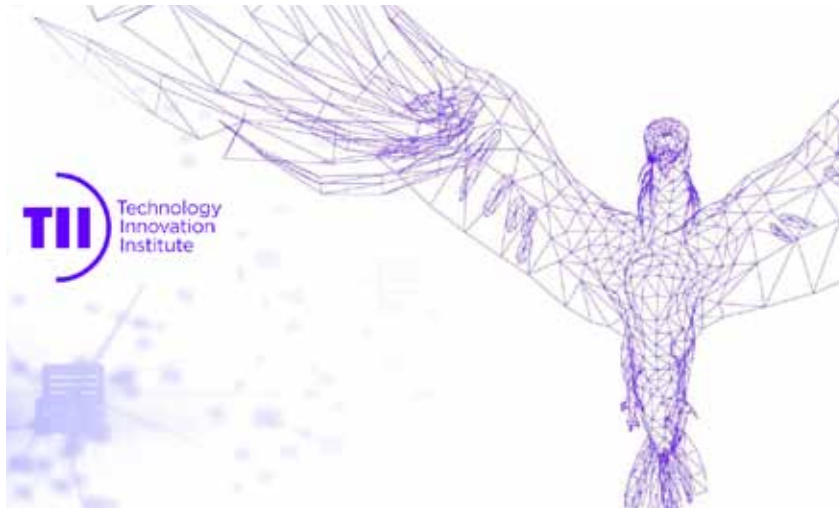


Falcon 180B: TII's new open large language model



The Technology Innovation Institute (TII) has unveiled Falcon 180B, an open source large language model that is making waves in the AI community. The new model boasts of a jaw-dropping 180 billion parameters and is trained on 3.5 trillion tokens. It currently reigns supreme on the Hugging Face leaderboard for pre-trained open large language models, and caters to both research and commercial use.

Falcon 180B stands shoulder to shoulder with giants like OpenAI's GPT-4 and Google's PaLM-2, despite being half their size. TII's commitment to high-quality training data shines through its meticulous approach.

This 360GB model is distributed with an Apache 2.0 license. It also has a chat version, is completely free and state-of-the-art.

A key strength is that it can handle diverse language tasks, including reasoning, coding, proficiency assessments, and knowledge testing. This is because it's trained on a rich and varied data set that includes web data, conversations, technical papers, and even a fraction of code. Hence, its performance in these tasks matches and often surpasses closed-source competitors like Meta's LLaMA 2.

This open-access model with 180 billion parameters empowers researchers and developers to explore new horizons in natural language processing, and opens the door to innovation across various domains, including healthcare, finance, education, and more.

In short, Falcon 180B's success underscores the value of open source initiatives in AI.

Quetzal-1 satellite project releases open source repository

The team behind the Quetzal-1 satellite project, initially launched in 2020 by students from Universidad Del Valle De Guatemala (UVG), has chosen to release an extensive open source repository, sharing its valuable insights and experience.

Quetzal-1, a CubeSat engineered by UVG students, achieved an impressive 211 days in orbit. Its mission objectives included gathering telemetry data and capturing black-and-white images of Earth. However, initial image quality suffered due to exposure issues.

EDB launches tool for easy Postgres cluster management

EnterpriseDB (EDB), a leader in enhancing Postgres for enterprise use, has unveiled its Trusted Postgres Architect (TPA) as an open source solution for PostgreSQL users. TPA simplifies the setup and management of high availability (HA) Postgres clusters, offering time-saving benefits and reducing the risk of errors.

Historically, HA cluster management has been complex, with potential impacts on performance and data accuracy. EDB aims to address this challenge and make HA clusters more accessible to organisations.

TPA offers a user-friendly structure, enabling enterprises to create and manage HA architectures effortlessly. This facilitates experimentation with cluster setups while maintaining optimal performance and reliability.

It features concise, declarative syntax, enabling users to experiment with trusted architectures for backups, replication, and protection against replication issues.

Combined with EDB Postgres Distributed's Always On architecture, TPA minimises downtime for active-active use cases. EDB backs TPA with its extensive Postgres expertise.

Jozef de Vries, chief product engineering officer of EDB, stated: "TPA embodies EDB's commitment to customer success and supporting the wider Postgres community. It reduces the chance of errors and saves time when deploying production database clusters."

EnterpriseDB's TPA allows organisations to confidently deploy and manage robust database clusters, enhancing operational resiliency and reducing errors in mission-critical applications.